



Optimization Sprint Report

TriNeura

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1. Data Exploration and Process Flow

- ❖ The dataset used in this sprint is the “Dementia Prediction Dataset”, which originally contained a mixture of medical and non-medical attributes. Since the hackathon requirement strictly specifies using only non-medical and self-reported information, all medical diagnostic variables were excluded during preprocessing.
- ❖ The target variable for this project is **DEMENTED**, representing whether an individual is identified as having dementia (**1 = Demented, 0 = Not Demented**). The initial dataset shape before preprocessing was **(3,664 rows × 1,024 columns)**. After removing invalid target entries, replacing encoded missing values, and selecting only non-medical relevant features, the final modeling dataset contained **3,664 records and 22 selected non-medical attributes** used for training and evaluation.
- ❖ The process flow followed in this project is summarized below:
 1. Loaded and inspected the dataset to understand feature structure and initial data distribution.
 2. Validated the target variable (DEMENTED) and removed records with invalid, missing, or non-numeric target values.
 3. Selected a subset of non-medical features based on the official data dictionary and domain relevance.
 4. Applied feature engineering to generate composite lifestyle and health risk attributes.
 5. Performed data cleaning and preprocessing, including handling missing values and converting all fields to numerical format.
 6. Split the dataset into training and testing partitions using an 80/20 stratified split to preserve class proportions.
 7. Trained four machine learning models (Logistic Regression, Random Forest, Gradient Boosting, and XGBoost) and evaluated performance using Stratified 5-Fold Cross-Validation with ROC-AUC as the scoring metric.
 8. Selected the best-performing model, Random Forest, for final evaluation and explainability analysis.
 9. Developed a simple dementia risk prediction function capable of estimating risk probability for new inputs.

2. Feature Engineering

2.1 Selected Non-Medical Features

- ❖ This phase focuses on selecting relevant non-medical attributes and creating meaningful derived features to enhance the model's ability to detect dementia risk from self-reported information only.

Category	Features
Demographic Factors	NACCAGE (Age), SEX, EDUC, MARISTAT, NACCLIVS, RESIDENC
Smoking History	TOBAC100, TOBAC30, SMOKYRS, PACKSPER, QUITSMOK
Alcohol Consumption	ALCOCCAS, ALCFREQ
Self-Reported Health Conditions	DIABETES, HYPERTEN, CVHATT, CBSTROKE, NACCSTYR

2.2 Feature Reduction Strategy

- ❖ Feature reduction was performed implicitly by excluding all medical diagnostic variables and retaining only the domain-informed non-medical variables.

2.3 Engineered Composite Features

Engineered Feature	Formula	Purpose
SMOKE_SCORE	$SMOKYRS \times PACKSPER$	Represents long-term smoking intensity
ALCOHOL_SCORE	$ALCOCCAS \times ALCFREQ$	Measures overall alcohol exposure load
HEALTH_RISK_COUNT	$DIABETES + HYPERTEN + CVHATT + CBSTROKE$	Aggregates cardiovascular comorbidities

❖ 2.4 Finalized Feature Set

The final feature set used for model training includes all selected non-medical variables plus the engineered features: SMOKE_SCORE, ALCOHOL_SCORE, and HEALTH_RISK_COUNT.

3. Data Preprocessing

- ❖ To prepare the dataset for effective model training, several preprocessing steps were applied. These transformations ensured data consistency, handled missing values appropriately, and preserved the predictive value of all relevant features.

- ❖ **Preprocessing Steps Applied**

Step	Description
Handling Encoded Missing Values	Special missing value codes such as -4, 88, 99, 999, 9999 were replaced with NaN to ensure correct interpretation as missing data.
Numeric Conversion	All selected features were converted to numeric types using <code>pd.to_numeric</code> to avoid type errors during model training.
Median Imputation	Missing values were imputed using median strategy via <code>SimpleImputer</code> , which is robust to extreme outliers and maintains feature distribution.
Stratified Train-Test Split	The dataset was split using an 80/20 stratified split, ensuring the same proportion of dementia vs. non-dementia cases in both sets.
Scaling (Applied selectively)	Standardization using <code>StandardScaler</code> was applied only to Logistic Regression, since tree-based models are scale-invariant.

- ❖ This preprocessing approach ensures fair model comparison and minimizes data leakage throughout the ML pipeline.

4. Model Building

- ❖ Multiple machine learning algorithms were developed and evaluated to identify the most effective model for dementia risk prediction using non-medical features.

Model	Description
Logistic Regression	Baseline linear classifier with <code>class_weight="balanced"</code> to address class imbalance.
Random Forest Classifier	Ensemble method of 400 decision trees with balanced class weighting and <code>min_samples_leaf=3</code> for regularization.
Gradient Boosting Classifier	Boosting approach using 400 estimators, <code>learning_rate=0.05</code> , and <code>max_depth=3</code> for controlled complexity.
XGBoost Classifier	High-performance gradient boosting variant using histogram optimization, 500 estimators, and <code>eval_metric="auc"</code> for ranking optimization.

❖ **Cross-Validation Strategy**

- ✓ A 5-Fold Stratified Cross-Validation method was used with ROC-AUC scoring to ensure reliable comparison between all models.

❖ **Hyperparameter Approach**

- ✓ No full grid or randomized search was required at this stage, but parameters were iteratively tuned based on cross-validation feedback, focusing on:
 - Tree count and leaf size (to prevent overfitting)
 - Learning rate adjustments
 - Class balancing to improve recall for dementia cases

5. Model Evaluation

❖ **Primary Evaluation Metric**

- ✓ Due to class imbalance, **ROC-AUC** was selected as the primary evaluation metric because it measures ranking performance independent of classification thresholds.

❖ **Model Performance Comparison**

Model	CV AUC	Test AUC
Random Forest	0.7833	0.7919
XGBoost	0.7526	0.7552
Gradient Boosting	0.7482	0.7504
Logistic Regression	0.7130	0.7157

- Best Performing Model: Random Forest Classifier
- Final Test ROC-AUC: 0.7919

❖ **Classification Report (Test Set)**

Class	Precision	Recall	F1-Score	Support
0 – Not Demented	0.84	0.78	0.81	27,522
1 – Demented / At Risk	0.55	0.64	0.59	11,518
Model Accuracy			0.74	39,040

Interpretation:

- The model demonstrates **strong performance** in correctly identifying non-dementia cases.
- Recall of **0.64** for the at-risk class is important because missing dementia risk cases can be more harmful than false alarms.

❖ Confusion Matrix

Actual/Predicted	Pred 0	Pred 1
Actual 0 (Not Demented)	21,541	5,981
Actual 1 (At Risk)	4,185	7,333

Interpretation

- **True Negative (21,541):** Correctly predicted not-demented cases
- **True Positive (7,333):** Successfully flagged at-risk individuals
- **False Negative (4,185):** Missed cases (opportunity for future improvement)
- **False Positive (5,981):** Acceptable trade-off since early detection prioritizes recall

6. Explainability & Model Interpretability

- ❖ Model explainability is essential in healthcare-related prediction systems, especially when identifying individuals who may be at risk of dementia. Since this study focuses on non-medical, self-reported information, interpretability helps validate whether the model learns meaningful, credible relationships rather than random patterns.

Explainability Techniques Used

Technique	Purpose
Feature Importance (Random Forest)	Measures how much each feature contributes to model decision-making.
SHAP (SHapley Additive Explanations) (recommended and applicable)	Provides individualized explanations showing how each feature influenced the risk prediction score.

- ❖ Random Forest was chosen as the final model partly because tree-based models offer strong explainability and are generally easier to interpret compared to deep neural architectures.

Tools / Technologies Used

Category	Technologies
Programming Language	Python 3
Data Processing	pandas, numpy
Visualization	matplotlib, seaborn
Machine Learning	scikit-learn, XGBoost
Explainability	SHAP (recommended), RF feature importance
Environment	Google Colab

Conclusion

- ❖ This study demonstrates that dementia risk can be predicted using only non-medical self-reported attributes, providing a promising approach for early, low-cost risk screening. Among four evaluated machine learning models, the **Random Forest classifier** achieved the highest performance with a **Test ROC-AUC of approximately 0.79**, showing strong ability to differentiate at-risk individuals. Feature importance analysis indicates that **age, education level, smoking and alcohol exposure, and cardiovascular health history** are highly influential predictors. Future improvements may include SHAP-based interpretability, external validation, and extended socio-behavioral feature integration for enhanced public health decision-support.